

A Crash Course in Arduino Programming

Lesson 4 Exercises

1. There are many Serial functions described in the Arduino documentation that we did not cover today. Pick one and create an example that uses it.
2. In the program we made for processing user input, we used a character array with a size of 32. What happens if you make the array too small to hold the messages you're sending?
3. In many cases, you will want users to select input commands from a certain range of numbers or from a specific collection of allowed words. Create a possible strategy for handling invalid user input.

Challenge Problems

1. Set up a circuit with at least 3 RGB LEDs. Write a program where, from the Serial monitor, the user can select which LED they want to change the color of and then input values for R, G, and B.
2. Create a program to control the direction and speed of a DC motor from the Serial monitor.
3. Explain how using the String() object can lead to memory fragmentation. How vulnerable is your research project to memory issues?